GPU-RESIDENT SHARED MEMORY

REMANON

"Maximum alignment, minimum detachment."

Nevine Fakhreddin

AMD Developer Hackathon: ACT II · Track 3 (Unicorn) · July 2026

github · linkedin · demo

Every agent starts from zero

The cost of multi-agent AI is dominated by re-processing context that was already processed.

62%

of total agent inference spend is redundant re-reading of already-processed context

Stanford Digital Economy Lab, 2026

250,000×

a 70,000-word document can expand beyond 100 GB of GPU state once a model reads it

Engram CTO, 2026

N ×

N agents on the same material means N private copies of it resident in GPU memory

duplication scales with team size

The bleed is not model quality. It is memory: the same knowledge, rebuilt and stored again for every mind.

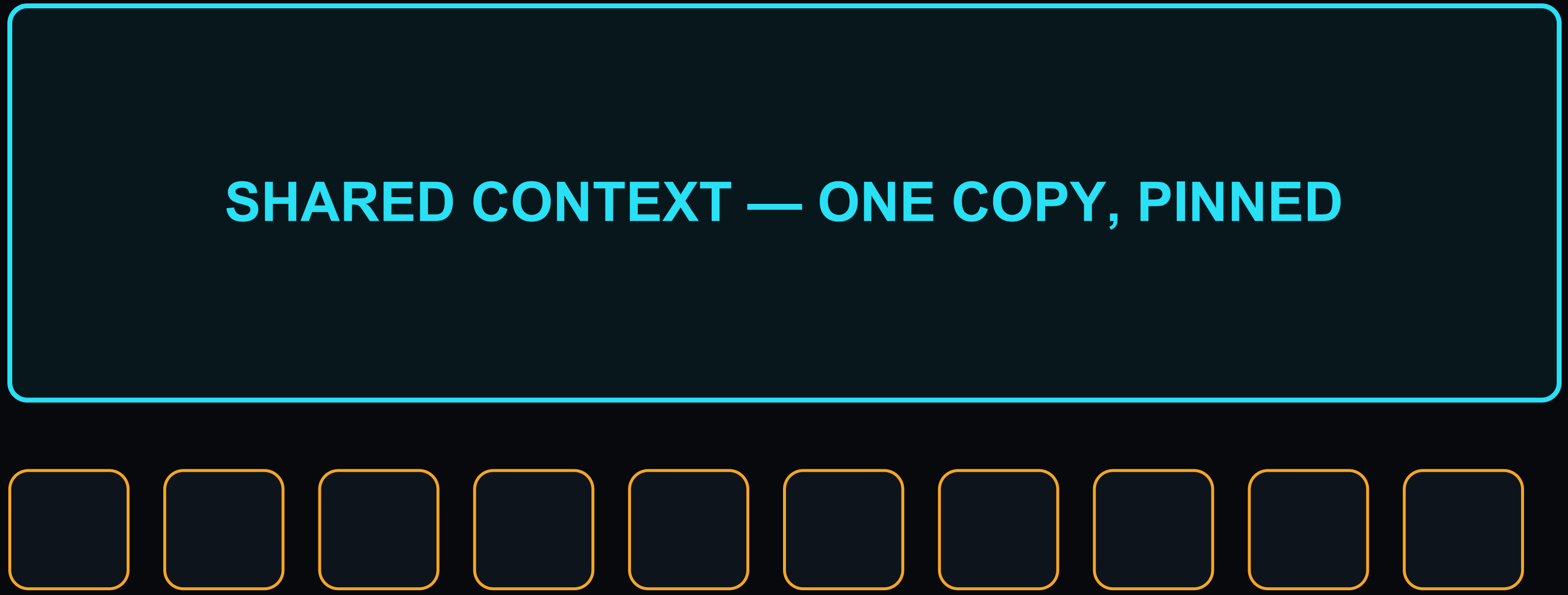
The same pages, paid for ten times

Ten agents reading one large context, today versus with Remanon.

TODAY

10 full copies resident in HBM. Memory and prefill paid 10 times.

WITH REMANON



Plus ten small private deltas. Each agent keeps only its own working notes.

One reading. One residence. Every mind served.

Strong solutions, at other layers

Each attacks real waste. None owns the GPU-runtime memory layer.

Prefix caching (vLLM)

Reuses identical prompt prefixes. Opportunistic and evictable; breaks when agents differ; no guarantee.

Retrieval (RAG)

Fetches the right passages, but every agent still re-prefills what it retrieved.

Learned memory — Engram, $98M

Compresses knowledge into weights. Powerful, but training-time and per-model.

Orchestration — Callosum, $10.25M

Routes workloads across chips. Scheduling, above the memory itself.

THE GAP Between weights and scheduling sits the GPU runtime, where context actually lives. No one guarantees shared memory residency there. That layer is Remanon's.

Remanon: the pinned-residency contract

From maybe cached, to guaranteed resident.

One living context block, resident in GPU memory, read concurrently by every mind on the team.

The name is the physics. *Remanence: the trace that persists in the material after the field is gone.*

Opportunistic caching (today)

- May be evicted at any moment
- Only identical prompt prefixes
- Per-engine, best-effort

Pinned residency (Remanon)

- Cannot be silently evicted — a contract
- Role-differentiated agents share the block
- One arbiter across all engines

Proven, not promised. 10 concurrent requests produce exactly ONE prefill, measured at the HTTP transport layer. 170 automated tests.

Three bands, two contracts — Lego by design

Every seam is a strict contract: the domain swaps above, the engines and hardware swap below.

BAND A · APPLICATION — incident root-cause analysis (swappable domain)

TRIAGE	CORRELATOR	HUNTER	TOPOLOGY	REPORTER

CONTRACT A · lease / materialize / generate (Remanon's own)

BAND B · REMANON CORE — memory arbiter: one copy per distinct model, pinned, never evicted

Engine Registry	Materializer	Residency Mgr	Budgeter	Metrics

CONTRACT B · OpenAI-compatible engine API (vLLM)

BAND C · SERVING & HARDWARE one vLLM engine per distinct model · ROCm · AMD Instinct HBM3

Inside the core — one memory, many minds

Five mechanisms, one guarantee.

1 Neutral master

Shared knowledge prefilled once, stored model-neutral. No model's private language.

2 Lazy materialization

One copy per distinct model, created only when that model first joins. Never per agent.

3 Pinned block

Materialized copies are pinned: no silent eviction, for the lifetime of the workload.

4 Delta budgets

Each agent's private context is small and budgeted around the block. Integer-byte admission law.

5 Cross-engine arbiter

One authority manages HBM across all engines. Memory becomes a governed resource.

Five minds, four models, one memory

The application layer: an incident-investigation team.

TRIAGE	CORRELATOR	HUNTER	TOPOLOGY	REPORTER
classify & route	cross-signal linking	log pattern search	dependency graph	RCA synthesis
gpt-oss-20b	gpt-oss-120b	Llama 3.3 70B	Qwen3-32B	gpt-oss-120b

Heterogeneous by design: the sharing unit is the model. The two cyan agents (Correlator, Reporter) draw from ONE pinned memory.

The demo domain: incident root-cause analysis

The core is domain-blind. A domain qualifies only if it passes two rules.

RULE 1 — SHARED

One reference body serves the whole team. In an incident, every investigator reads the same telemetry, topology, and history.

RULE 2 — LONG & STABLE

The context is large and reused enough that keeping it resident beats re-loading it. Enterprise telemetry is exactly that.

THE DATA

- Loghub — the peer-reviewed benchmark for log-analysis research
- Real production failures with published ground truth, not synthetic data we could tailor
- Nearly one million real log records ingested across HDFS, BGL, Hadoop, Spark, and Thunderbird
- Anomaly thresholds profiled FROM the data itself via SQL, independent of replay speed

Memory is the hero — fits or does not fit

AMD's advantage here is absolute, not a percentage.

192 GB HBM3, fused on one AMD Instinct MI300X

5.3 TB/s memory bandwidth: every agent reads the shared block at full speed

80 GB CARD

Four models plus the resident shared block exceed capacity. The architecture cannot even load. The idea dies at the door.

MI300X — 192 GB

Weights plus the pinned shared block plus many agents, resident together with visible headroom. Not faster: possible.

Mechanism proven live on a 48 GB AMD card, where exactly one of the four models fits: the very scarcity Remanon's admission law governs.

Two versions, honestly separated

What actually ran on the available hardware, and what the full system computes to.

MEASURED REALITY

what actually ran on AMD

- One model (gpt-oss-20b) served live on AMD gfx1100
- Real flood of production logs, real classified verdicts
- 46 of 48 GB used · 14.3 GiB weights · 26.67 GiB cache
- 5.8× reuse speedup · 66.3% hit rate · downloadable ledger

RESEARCH VISION

the full system on the right hardware

- Five agents, four models, one pinned master each
- Resident together on 192 GB HBM3
- Grounded in the measured mechanism plus 170 tests
- Labelled COMPUTED throughout, never as measured

Why only one model ran live. The four models need about 162 GB and physically cannot co-reside on a 48 GB card. So the mechanism is measured on what fits, and projected by transparent arithmetic to the hardware it was designed for.

Validated on AMD — every number labelled

Captured live on AMD gfx1100, 48 GB, serving gpt-oss-20b under vLLM.

GPU / architecture	**AMD gfx1100 · 48 GB VRAM**	MEASURED
Model weights resident	**14.3 GiB**	MEASURED
Key-value cache available	**26.67 GiB · 582,576 tokens**	MEASURED
VRAM used while serving	**46.02 GB of 48 GB**	MEASURED
Shared-context reuse	**cold 10.635 s → warm 1.839 s = 5.8× faster**	MEASURED
Engine prefix-cache hit rate	**66.3%**	MEASURED
Prefills avoided across the run	**110**	MEASURED
Evictions of a pinned master	**0**	MEASURED

Two independent signals confirm the same fact: our timed 5.8× and the engine's 66.3% hit-rate counter. Logic proven by 170 automated tests, including transport-layer contract tests. Real code, not slides.

Engineering discipline

Built solo, with big-tech gates from day one.

170

automated tests: contract, concurrency, cross-layer invariants

0

leaked leases across every full run. Accounting closes exactly

1

prefill per model under 10 concurrent requests, transport-measured

$10^9\times$

replay speed with identical detection. Burst logic is time-true

- CI and pre-commit on every commit, one pinned toolchain
- Observation plane read-only by construction: zero mutating endpoints
- No code path exists that can evict a pinned master: correctness by construction, not by policy

The market just validated the problem

Two funded companies attack agent-memory waste, from opposite sides.

Engram

$98M raised

June 2026 · Sequoia, Kleiner, Karpathy · $600M valuation

Learned memory baked into model weights. Training-time, per-customer.

REMANON

GPU-runtime residency layer

the layer both sides must pass through

Guaranteed shared residency inside the GPU, across different models, at inference time. The gap between them.

Callosum

$10.25M raised

Feb 2026 · Plural + UK ARIA

Multi-model orchestration across chips. Scheduling layer, AMD-adjacent.

Top-tier investors priced this problem at nine figures this year. Remanon answers it where neither competitor operates.

The business case

Token bills are the new cloud bills, and memory is where they leak.

COST PER MULTI-MODEL TEAM

	Without Remanon	With Remanon
GPUs needed	several 80 GB cards	one MI300X 192 GB
Why	162 GB does not fit 80 GB	the team fits with headroom
Hourly cost	multiplied per card	a single card's rate
Saving	High	a substantial reduction

WHO PAYS

- Platform teams running agent fleets
- GPU clouds selling large-memory silicon
- Enterprises adopting agents at scale

WHAT THEY BUY

- Collapse of redundant prefill spend
- More agents per GPU: capacity, not just savings
- A runtime layer: per-GPU licensing, no retraining

Measured: on real 48 GB hardware, Triage alone avoided 110 prefills and saved 10 GB against a per-agent baseline.

The research track

The engineering opens three publishable problems, the seed of a paper.

1

Cross-model materialization

How far can one neutral context be shared across incompatible cache geometries, and at what fidelity cost?

2

Residency arbitration

What is the optimal pinning policy when block, deltas, and headroom compete under live load?

3

Contract formalization

Can no-eviction residency be specified and verified as a formal memory contract for agent runtimes?

Three editions of the master documentation — business, technical, research — grow from one architecture.

Live operations theater

One deployed site. Two honest paths. A recorded replay of a real run.

MEASURED REALITY

The live single-model run on AMD gfx1100. A storm of real records drifts in, Triage classifies each case, and the full ledger exports as CSV and Markdown: the artifact an on-call engineer actually keeps.

RESEARCH VISION

The full four-model city on 192 GB, with an interactive memory allocator that computes, in real arithmetic on real model sizes, exactly when a team fits an 80 GB card, when it fits 192 GB, and when it overflows.

Visit it live: **nevineakf.github.io/remanon**

One system. One memory. Many minds.

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